

Are armband sEMG devices dense enough for long-term use?—Sensor placement shifts cause significant reduction in recognition accuracy

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ABSTRACT

Myoelectric control systems (MCSs), which recognize motions through surface electromyograms (sEMGs), present potential applicability for clinical, recreational, and motion-assisting purposes. To increase the adoption of armband device-based MCSs, the performance of motion recognition algorithms should be determined over long periods and sensor placement shifts. We prepared an sEMG dataset to assess motion recognition algorithms for practical use over long periods with varying sensor placement. The dataset comprises 30 recording sessions over 40–42 days, in which sensors were placed at three different placements. We used an armband eight-channel sEMG device for capturing 22 types of forearm motions from five healthy male subjects. To consider only motion periods to learn classifiers, we extracted relevant 1.5-s segments via multiscale sample entropy. We evaluated the dataset on a conventional motion recognition algorithm, finding robust intraday performance but significantly deteriorated inter-day performance under varying sensor placement. Hence, the armband sEMG device is dense enough for short-term use but not apt for long-term use regarding the conventional recognition algorithm. Adaptation techniques are required for developing armband device-based MCSs for long-term use. The dataset and sample codes from this study are publicly available at [GitHub](https://github.com).

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1. Introduction

Myoelectric control systems (MCSs) translate surface electromyograms (sEMGs) during specific motions into predefined commands and constitute intuitive interfaces for prostheses, virtual reality, and robotic applications [1–3]. Although simple thresholding in commercial prostheses is the mainstream control technique for MCSs [4], machine learning algorithms are being increasingly adopted for accurate, fast, and stable motion recognition [5–9]. The development and evaluation of such algorithms have been widely conducted using publicly available sEMG datasets such as the myoelectric control development toolbox [10] and NinaPro database [11,12]. The resulting motion recognition algorithms and techniques can benefit society given their wide applicability.

Modern wearable and wireless armband sEMG devices are attracting research interest because they are easy to use, affordable, and comfortable, being highly suitable for MCSs [13–15]. The device design and development are based on high-density sEMG

recording and the data analysis for finding a small number of bipolar electrode pairs equally spaced around the forearm to capture enough information for motion recognition [16,17]. For instance, the Myo gesture control armband, which is an eight-channel sEMG device, has been used in several studies involving various applications such as prosthetic hands [18], virtual reality [19], and robot teleoperation [20].

Nevertheless, armband sEMG devices present poor placement repeatability during reattachment. This problem is particularly noticeable during long-term use to perform activities of daily living, as the user removes the device after MCS use and reattaches it before using it again. Reattachment inevitably leads to varying electrode placement that affects the distribution of sEMG signals and may undermine the performance of underlying motion recognition algorithms. Moreover, as machine learning relies on training data to predict outcomes for unseen data, motion recognition using this approach can be unstable and inaccurate if the variability of aspects such as placement between training and testing datasets and real-time use are neglected. However, existing datasets assume sEMG data as being captured from the same placement regardless of the measurement periods, which can span over several days, and sensors for acquisition, thus neglecting variability. To the best of our knowledge, no study on armband sEMG motion recognition has

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systematically analyzed the performance of conventional machine learning considering varying sensor placement.

In this study, we assessed the performance of sEMG recognition algorithms over a long period of use and with shifts in sensor placement. In addition, we analyzed the performance of sEMG recognition over short periods to confirm whether the small number of bipolar electrode pairs equally spaced around the forearm can actually capture enough information. To achieve these objectives, we acquired an sEMG dataset using the Myo gesture control armband over 40–42 days (30 recording sessions) at three different sensor placements (neutral position, eight-millimeter anticlockwise rotation, and eight-millimeter clockwise rotation). The dataset has the following characteristics: (i) long-term sEMG recordings (sessions over 30 days), (ii) varying sensor placement (three placements), (iii) diversity of motions (22 types of motions comprising four-degree-of-freedom (DOF) forearm motions, with each type being performed four times per day), (iv) motion onset detected by sample entropy, and (v) effective 1.5-s segments comprising the motion events. These characteristics allow the study of machine learning techniques under realistic conditions of regular long-term MCS use. In our previous study [21], we used a similar dataset containing data from two subjects for assessing the effectiveness of transfer learning to mitigate MCS instability during long-term use. In the present study, we achieved the following novel contributions: (i) number of subjects increased from two to five, (ii) motion labeling to train classifiers, (iii) segmentation to use only motion periods during training and performance evaluation, (iv) investigation of intraday and inter-day data quality with multiple initial settings, and (v) preparation of dataset with sample codes publicly available on a GitHub repository.

2. Materials and methods

2.1. Participants

Five healthy subjects (male, right-handed, aged 24–28 years, mean age of 25.4 years) participated in this study. Two subjects (S01 and S02) had previous experience with MCS operation, and the others (S03–S05) were naïve. Their right forearm girths, where the armband sEMG device was attached, were 27.8 cm for S01, 24.9 cm for S02, 26.5 cm for S03, 26.1 cm for S04, and 25.0 cm for S05. Each subject provided informed consent before beginning the experiment on each day of the study. The study was approved by the Institutional Review Board from the National Institute of Advanced Industrial Science and Technology (AIST), Japan (HF2017-784).

2.2. Armband sEMG device and placement

We acquired eight-channel sEMG signals from the right upper limb of the subjects using the Myo gesture control armband (Thalmic Labs, Kitchener, Canada). The signals were sampled at 200 Hz and high-pass filtered at 15 Hz through a fifth-order Butterworth filter. The recordings per subject took place for 30 consecutive days excluding weekends (Saturday and Sunday), corresponding to a study period of approximately 1.4 months (40–42 days).

The sensor placement for each day was selected to be one of the following types:

1. Neutral position (N),
2. Eight-millimeter anticlockwise inward rotation (I), and
3. Eight-millimeter clockwise outward rotation (O),

where references inward/outward and anticlockwise/clockwise were considered with respect to the subject's viewpoint (Fig. 1). The

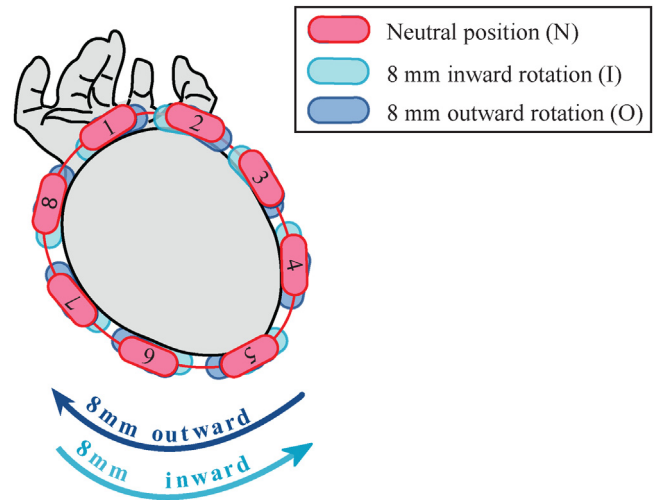


Fig. 1. Myo gesture control armband mounted on neutral position, 8 mm inward rotation, and 8 mm outward rotation modified from [22].

Table 1

Sensor placement per day of study (N, neutral position; I, eight-millimeter anticlockwise inward rotation; O, eight-millimeter clockwise outward rotation).

Day	Pos.	Day	Pos.	Day	Pos.
1	N	11	O	21	O
2	I	12	N	22	I
3	O	13	N	23	O
4	N	14	O	24	I
5	I	15	O	25	I
6	I	16	I	26	N
7	O	17	I	27	N
8	O	18	I	28	I
9	N	19	N	29	N
10	N	20	O	30	O

change in data distribution was prevented by setting the maximum distance between placements to less than 2 cm [23], resembling the user attempting to repeat the same sensor placement every day for MCS use. However, we selected the 8 mm rotations because recent studies have simulated the effects of doffing and donning prostheses via lateral displacement of electrodes by 8 mm and reported large changes of sEMG data distributions that make the performance of classifiers decreased [24,25]. Measurements from each placement were acquired 10 times over the 30 days (see Table 1). The sensor placement permutation was initialized on the first day for S01 according to a uniform distribution and shared with all subjects. For placement N, the fourth channel containing a Bluetooth communication port was placed 1 cm distal to the belly of the brachioradialis muscle, and the seventh channel was placed 6 cm distal from the olecranon. The other six channels were equally spaced around the forearm. The three sensor placements were marked on each subject's forearm using permanent ink. We checked the marks on each day of the study and carefully remarked the fading ink.

2.3. Experimental procedure

During the experiments, each subject stood in front of a laptop PC in a relaxed posture and put both forearms approximately 10 cm forward to avoid contact with objects when performing the requested hand motions. The hand palms remained facing each other. To intuitively instruct the target motions to the subjects, we implemented a 3D graphical user interface based on C++ bindings from the Myo software development kit [26] and the Unity platform (Unity Technologies, San Francisco, CA, USA).

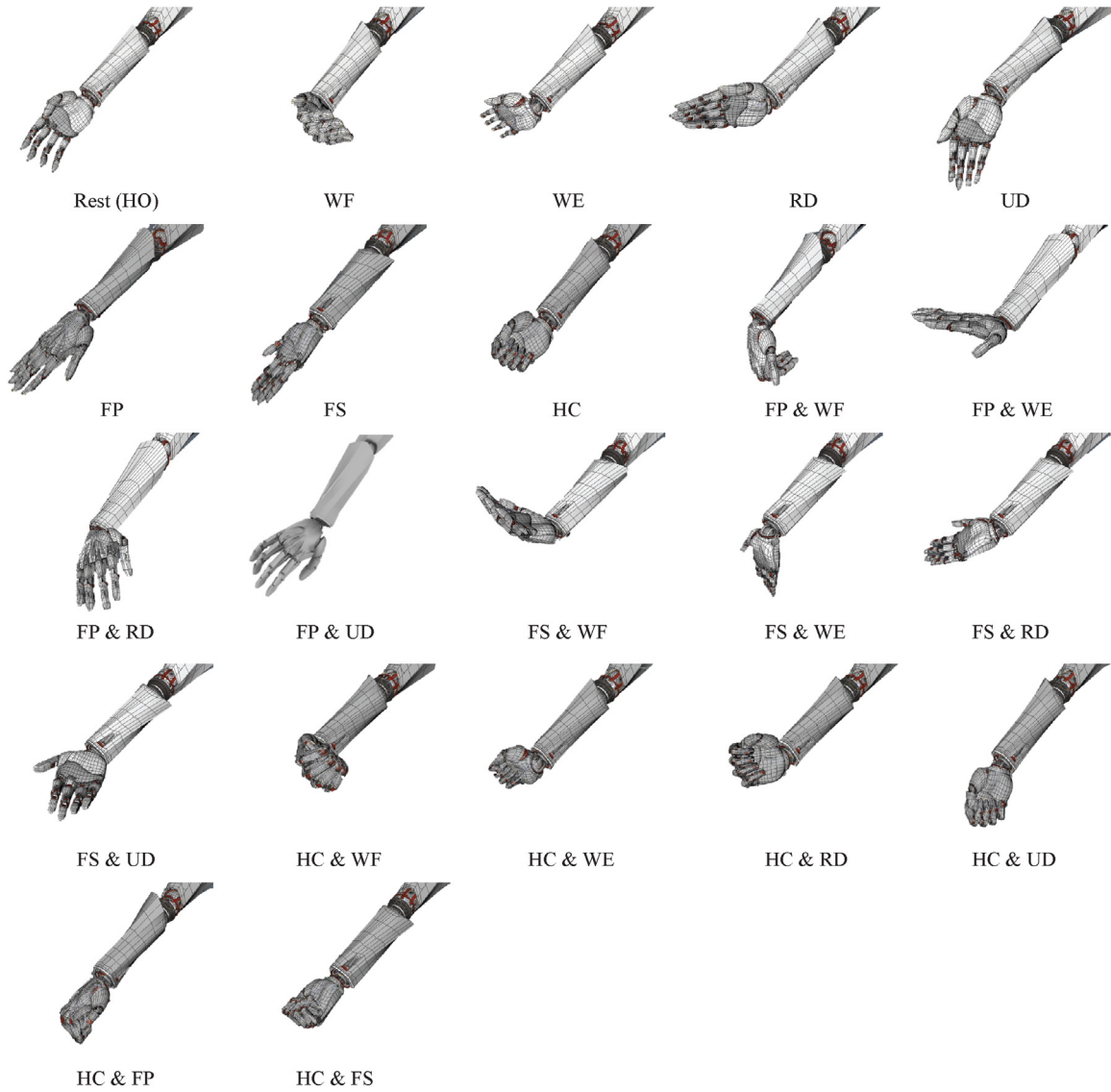


Fig. 2. Twenty-two types of motion combinations.

We designed 22 types of mirror motions to be performed by each subject based on simultaneous and proportionally controlled four-DOF manipulation [27]. Each motion was designed as a combination of the following eight fundamental gestures: (i) hand opening (HO), (ii) wrist flexion (WF), (iii) wrist extension (WE), (iv) radial deviation (RD), (v) ulnar deviation (UD), (vi) forearm pronation (FP), (vii) forearm supination (FS), and (viii) hand closing (HC). Movies of each fundamental gesture can be found in the GitHub repository of this study (under *hand motions* folder). In addition, the motions are visualized in Fig. 2.

Every day of the experiment, the subject repeated the 22 motions four times in the same order, resulting in 88 trials per session. Each trial lasted 6 s from $t=0$ to 6 s. At $t=1$ s, the 3D forearm model in the graphical user interface moved from the resting state to execute a gesture for 2 s ($t=1$ to 3 s). The subject followed the target motion (after the reaction time) and maintained the final posture (Fig. 3). Then, the subject returned his forearms to the initial position within 1 s ($t=3$ to 4 s) and adopted a relaxed posture for 2 s ($t=4$ to 6 s). Each subject rested for 2–5 min after performing the last single-DOF (eighth) motion. The eight-channel sEMG signals from the 88 trials were acquired each experiment day, resulting in 2640 trials over the 30 days of the study.

2.4. Data segmentation

To use only motion data from the 6-second trials, we extracted segments of 1.5 s (i.e., 300 sampling points) starting from the motion onset, which was detected via multiscale sample entropy [28]. Sample entropy is similar to approximate entropy but less biased measure than it because sample entropy does not count a self-match, is independent of data length, and demonstrates relative consistency, resulting in precisely quantifying the regularity and complexity of data [29,30]. In addition, we used an expanded measure of sample entropy called multiscale sample entropy, which effectively emphasizes the timing of sudden sEMG signal fluctuations generated by muscle contraction [31,32]. It considers the following parameters: (i) window length N_w , (ii) shifting size N_s , (iii) maximum length l of data for matching templates, and (iv) tolerance or slope r of sigmoid function. In [32,33], parameters (N_w, N_s, l, r) were set to (40, 5, 2, $0.25 \times SD$), where SD is the standard deviation of the evaluated signal.

For motion onset detection, we applied max normalization to the originally calculated sample entropies and experimentally determined a general threshold of 0.4. If the normalized sample entropy of the channel data initially exceeded the threshold, the

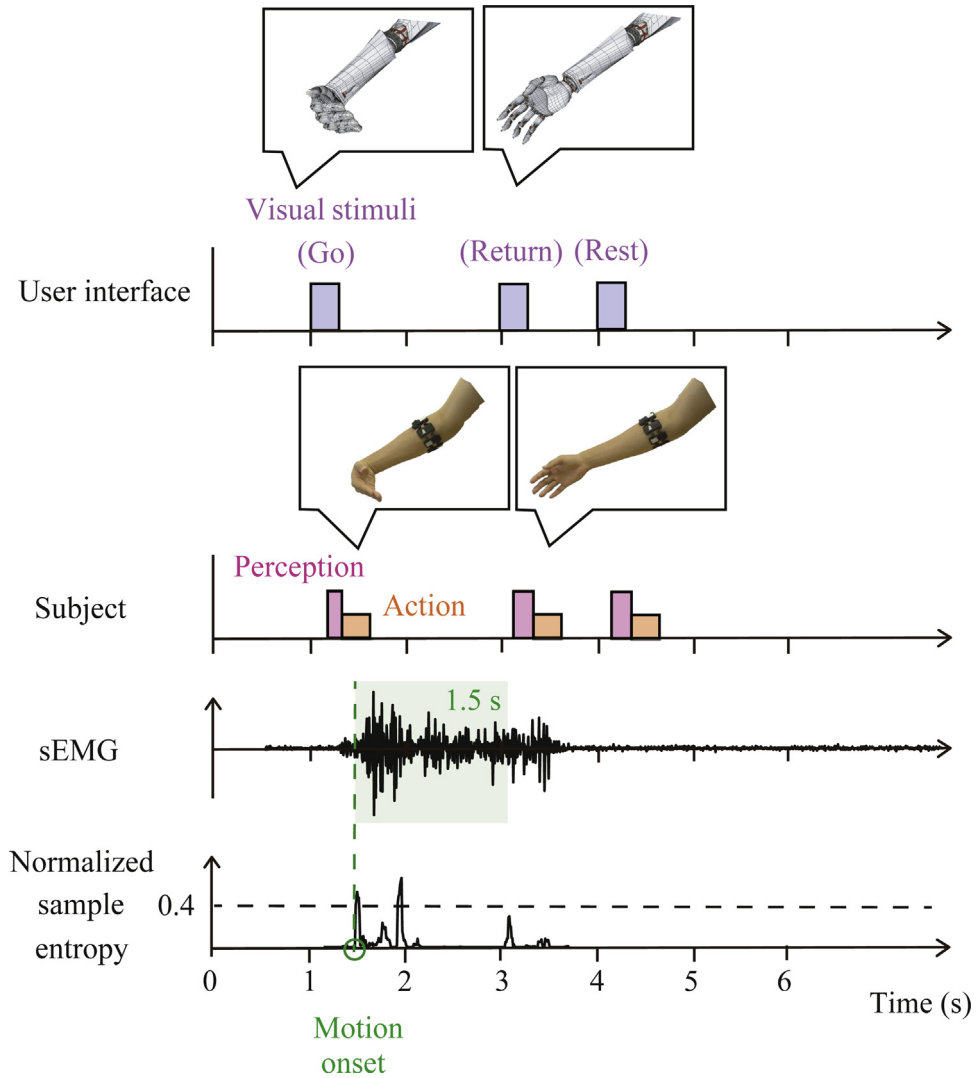


Fig. 3. Computerized motion instruction for wrist flexion, motion onset detection, and gesture execution sequence in one trial.

corresponding time was identified as candidate instant for motion onset. The first point among eight candidates exceeding the threshold in a motion trial was considered to be the onset instant, as illustrated in Fig. 3.

2.5. Feature extraction

Each 1.5-s sEMG segment was further split into analysis windows of 250 ms ($T=50$) every 50 ms (4/5 overlap) with the predefined class labels. The window length should remain below 300 ms to ensure a perceivable response time by the user during MCS operation [34].

Time-domain features combined with autoregressive (AR) model coefficients are the gold standard for sEMG data processing [35]. The features are composed of 11-dimensional features: (i) mean absolute value (MAV), (ii) zero crossing (ZC), (iii) slope sign change (SSC), (iv) waveform length (WL), (v) root-mean-square (RMS) value, and (vi–xi) sixth-order AR coefficients. Hence, the feature dimensionality per analysis window was 88 (11 dimensions $\times$ 8 channels) with the feature definitions per channel listed in Table 2.

The histograms of five time-domain features (MAV, ZC, SSC, WL, and RMS value) from the three sensor placements were separately computed for assessing their effect on data distribution. We chose

Table 2

Feature definitions per channel.

Feature	Definition per channel
MAV	$f = \frac{1}{T} \sum_{t=1}^T x_t $
ZC	$\{x_t > 0 \text{ and } x_{t+1} < 0\}$ or $\{x_t < 0 \text{ and } x_{t+1} > 0\}$ $ x_t - x_{t+1} \geq 0.1$
SSC	$\{x_t > x_{t-1} \text{ and } x_t > x_{t+1}\}$ or $\{x_t < x_{t-1} \text{ and } x_t < x_{t+1}\}$ $ x_t - x_{t+1} \geq 0.1$ or $ x_t - x_{t-1} \geq 0.1$
WL	$f = \sum_{t=1}^{T-1} x_t - x_{t+1} $
RMS value	$f = \sqrt{\frac{1}{T} \sum_{t=1}^T x_t^2}$
AR coeff.	$x_t = -\sum_{i=1}^6 f_i x_{t-i} + e_t$

the data of the fourth channel for visualization as it was the reference for sensor attachment, and thus its position was the most strictly regulated throughout the study.

Table 3

Motions and labels for training classifiers (HO, hand opening; WF, wrist flexion; WE, wrist extension; RD, radial deviation; UD, ulnar deviation; FP, forearm pronation; FS, forearm supination; HC, hand closing). The labels for each model represent independent opposing motions in a single DOF (labels ± 1) defined by musculoskeletal constraints of the right forearm or resting state for the DOF (label 0). Motion data with label ‘-’ were not used to train the classifier.

Motion index	Motion	Model				
		1	2	3	4	5
1	Rest (HO)	-1	0	0	0	-1
2	WF	1	1	0	0	-
3	WE	1	-1	0	0	-
4	RD	1	0	1	0	-
5	UD	1	0	-1	0	-
6	FP	1	0	0	1	-
7	FS	1	0	0	-1	-
8	HC	1	0	0	0	1
9	FP & WF	1	1	0	1	-
10	FP & WE	1	-1	0	1	-
11	FP & RD	1	0	1	1	-
12	FP & UD	1	0	-1	1	-
13	FS & WF	1	1	0	-1	-
14	FS & WE	1	-1	0	-1	-
15	FS & RD	1	0	1	-1	-
16	FS & UD	1	0	-1	-1	-
17	HC & WF	1	1	0	0	1
18	HC & WE	1	-1	0	0	1
19	HC & RD	1	0	1	0	1
20	HC & UD	1	0	-1	0	1
21	HC & FP	1	0	0	1	1
22	HC & FS	1	0	0	-1	1

2.6. Classifier

We designed five linear models that can be directly adapted to a real-time 3D graphics application controlling the four-DOF forearm motions. The first model discriminates whether the forearm realizes a specific motion, acting as trigger for an MCS, denoted as model 1 in Table 3. It uses the first motion index (rest) as class label -1, because the motion only has the sEMG information during resting (no motion). The other four models were designed as follows. Each motion was composed of four independent opposing movements defined by the musculoskeletal constraints of the right forearm (i.e., limit of four DOFs). These patterns are denoted as models 2–5 in Table 3, where the unassigned motions in model 5 indicate that the data from those motions were not used to construct the model, which cannot distinguish the degree of hand aperture but can discriminate the state of the hand as being either open or closed. In addition, class label 0 in models 2–4 can provide commands for realizing both single- and multiple-DOF motions. If models 2–5 show a predicted label set (0, 1, -1, 1), the motion can be detected as motion “RD&FS&HC” (Fig. 4). While each model handles only two class labels (a motion or its opposite), realizing a DOF-changeable interface is unfeasible.

To simultaneously identify the five patterns, we constructed linear discriminant analysis (LDA) classifiers ($m = 1, \dots, 5$), which are considered the gold standard for sEMG-based motion recognition [36–38], and predicted class label $y \in \{\pm 1\}$ or $y \in \{-1, 0, 1\}$ from the 88-dimensional feature vector, $\mathbf{f}$, described in Section 2.5:

$$y = \mathbf{w}_m^T \mathbf{f} + \eta, \quad (1)$$

where $\mathbf{w}_m$ is a coefficient vector for model m and η represents noise. Given a new feature vector $\mathbf{f}$, the class label was predicted as

$$\hat{y} = \text{sign}(\mathbf{w}_m^T \mathbf{f}'). \quad (2)$$

2.7. Validation

To examine the discriminability obtained from our long-term dataset under the various data distributions, we prepared three evaluations, as detailed below. For each evaluation, we constructed the corresponding LDA classifiers and determined the recognition performance per subject.

2.7.1. Intraday evaluation

The intraday data quality was verified via leave-one-out cross-validation applied to the trials to clarify whether the Myo armband can capture sEMG information to suitably recognize combined forearm motions. If the data quality were poor, noisy and undistinguishable data could have contributed to increased error. The four-trial data of target day $s \in \{1, \dots, 30\}$ were separated into training and validation datasets in each step of cross-validation. Each validation dataset contained 22 1.5-s segments corresponding to the different motions, whereas the other 66 segments were further split into training data and nonintervention data to verify the learning ability of the classifiers.

2.7.2. Inter-day evaluation with fixed training data

To assess the inter-day data quality over the 30 sessions, we first determined the motion recognition performance using a fixed training dataset. Specifically, the sEMG data from the first day of each sensor placement (N, I, or O) was used for training, and the recognition performances on any target day $s' \in \{2, \dots, 30\}$, $\{3, \dots, 30\}$, or $\{4, \dots, 30\}$ were obtained. In addition, the effect of sensor placement was assessed to determine the variant retrieving the highest deterioration of recognition performance. For a fair evaluation, we performed a modified leave-one-out cross-validation during the trials. Specifically, the first-day data of the sensor placement consisting of 88 segments were always used for training, and validation data of a target day were cross-validated within that day.

2.7.3. Inter-day evaluation with incremental training data

This evaluation was equal to that using fixed training data, except for the recognition performance on a target day. Specifically, the training data included all data from the preceding days since the first day of sensor placement (e.g., recognition performance from the 25th day considered the data from the preceding 22–24 days for training).

2.8. Performance evaluation

The error in motion recognition was defined for each model as

$$\text{Error} = \left(1 - \frac{1}{M} \sum_{m=1}^M \frac{N_{\text{correct}}^{(m)}}{N_{\text{total}}^{(m)}} \right) \times 100\%, \quad (3)$$

where $N_{\text{correct}}^{(m)}$ is the number of correct predictions and $N_{\text{total}}^{(m)}$ is the total number of datapoints.

We obtained the confusion matrices expressed in percentages to compare the performance among models. Each element of the matrices represents the amount of correct (diagonal elements) and incorrect (off-diagonal elements) predictions per model.

3. Results

3.1. Histograms of normalized features

Fig. 5 shows the histograms of the five time-domain features from the fourth sEMG channel containing the 22 motion types with a kernel smoothing function fit. The histograms were constructed using 30 bins. In all the features except for WL, the outward and

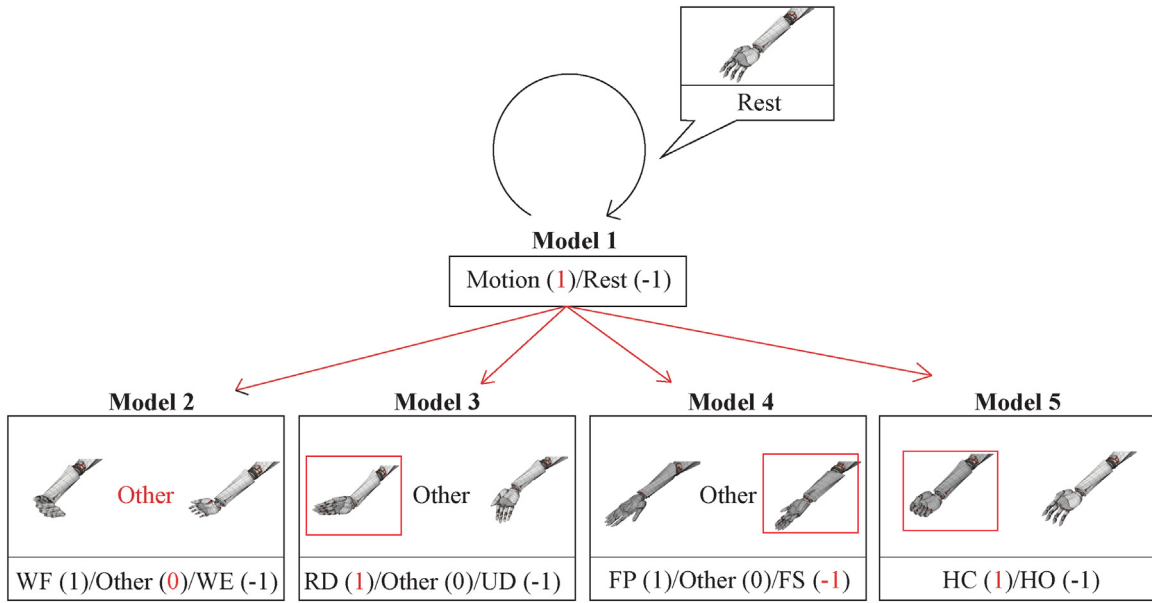


Fig. 4. Two-step motion recognition procedure for controlling four-DOF forearm manipulation including busy/idle states.

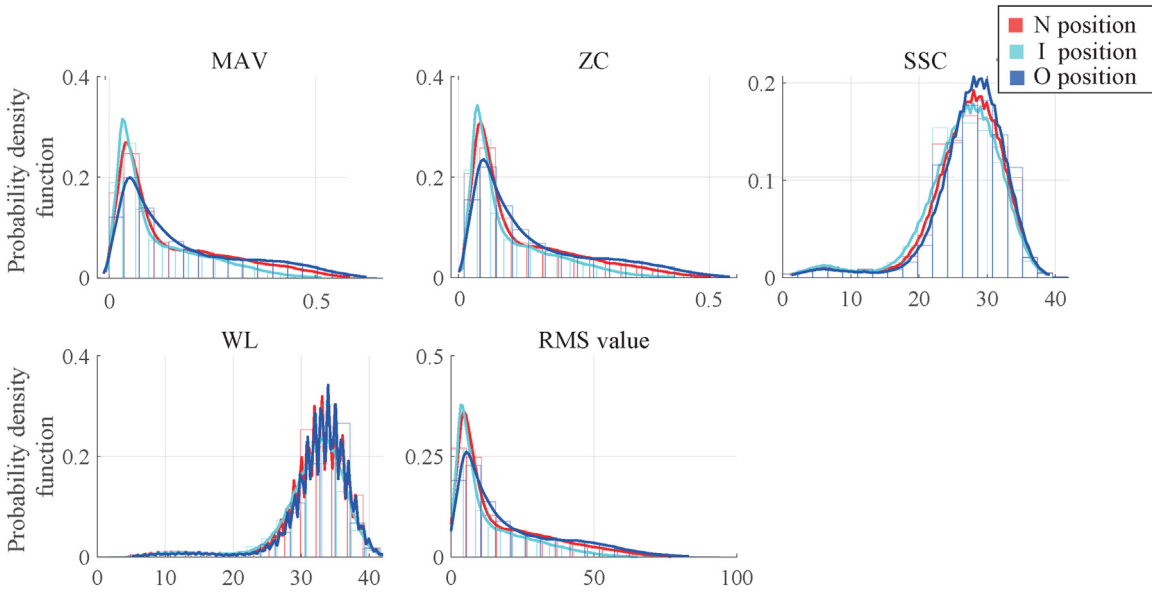


Fig. 5. Histograms with a kernel smoothing function fit of time-domain fourth-channel features at three different sensor placements.

inward rotations of the sensors from the neutral position changed the centers of distributions to the right and left, respectively. This phenomenon means that the 8 mm inward rotation involves moving away from the target (brachioradialis) muscle. Although the 8 mm outward rotation also involves moving away from the muscle, the fourth channel captured stronger features in this placement than in the neutral position. This might be caused by crosstalk, in which the channel captures additional information from neighboring muscles, such as the extensor carpi radialis longus muscle, after the 8 mm outward rotation. The probability distributions of the features obtained from the three sensor positions over 10 days are similar. However, these trends differ from a previous study [39], except for MAV (see Figure 2 in [39]). The trend divergence is possibly caused by the following differences in the studies: (i) number of subjects (8 vs. 5), (ii) number of motion types (9 vs. 22), (iii) placement of reference channel (2 cm distal to the elbow vs. 6 cm distal from the olecranon), and (iv) measurement period (1 vs. 10 days).

3.2. Intraday motion recognition

The intraday recognition errors obtained from leave-one-out cross-validation over trials are shown in Fig. 6. For all subjects and days, increasing the number of training trials improved prediction, with the lowest error obtained for three training trials. The errors ranges were 5.6–21.8%, 4.4–17.7%, and 3.8–17.5% for one, two, and three training trials, respectively. A two-way repeated measures analysis of variance (ANOVA) using the recognition errors of all subjects and days showed a statistically significant effect of the number of training trials ($F(2, 6741)=41.8, p<0.001$) but no significant effect of sensor placement ($F(2, 6741)=1.31, p=0.269$). A post-hoc analysis using Tukey test revealed significant differences ($p<0.001$ and $p=0.0476$) when consecutively increasing the number of training trials (i.e., from one to two and from two to three training trials). In addition, no significant interaction was observed between the number of training trials and sensor placement ($F(4,$

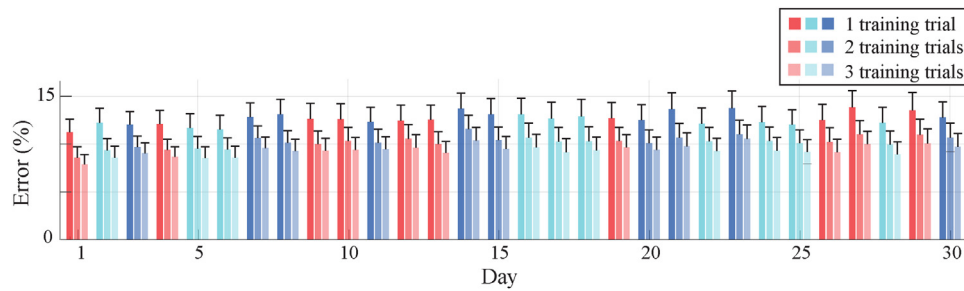


Fig. 6. Grand means and standard errors of intraday motion recognition errors. In each three-bar group, the rightmost bar indicates the error using the highest amount of training data for prediction.

6741) = 0.019, $p = 0.999$). Thus, the learning ability of classifiers was consistent among sensor placements for three training trials.

The performances were close to the results of previous studies (error of $8.3 \pm 6.8\%$ in [39] and 18.0% in [8]). These studies used different datasets (in-house dataset and exercise B of DB 5-2 in NinaPro database [11,12]) but the same Myo gesture control armband for sEMG measurements. Although our dataset and other databases have some differences such as number of trials (4 for ours vs. 5 for [39] vs. 6 for [8]) and motion types (8 vs. 9 vs. 9), all datasets contain similar configurations, and thus our recognition approach can be regarded as having baseline accuracy.

The accumulated confusion matrices expressed in percentages for the five models obtained from data on the respective days are shown in Fig. 7. The matrices based on one to three training trials contain data from the five subjects. The high values of diagonal elements indicate that the models accurately predicted the target class labels. Models 1 and 5, which predict two classes, namely, resting/hand closing (−1) and moving/hand opening (1), almost achieved 100% accuracy regardless of the number of training trials. These results support those from previous studies regardless of using clinical sensors [40,41] or the armband device [42]. The results for models 1 and 5 suggest that our dataset provides a classification quality equal to that of data from previous studies. On the other hand, model 4, which predicted three classes, namely, forearm supination (−1), other (0), and forearm pronation (1), exhibited the lowest performance (60.0–79.3%). Generally, it is difficult to identify forearm pronation/supination from sEMG signals [43,44], because these motions activate two pronator muscles (pronator teres and pronator quadratus) and two supinator muscles (supinator and biceps), which arise from different bones and are close to the bones [45]. Therefore, the corresponding sEMG signals present a relatively low amplitude and complex activity. The results from model 4 suggest that our dataset has the same limitation regarding classification of pronation and supination.

To understand classification errors in real-time MCS operation, we prepared a movie named `online processing`, whose link can be found in file `Readme.md` from our GitHub repository. The movie shows MCS control of a 3D graph using recognition outputs under the same processing parameters as described in the previous section (i.e., 250 ms analysis window updated every 50 ms). The average motion recognition error for this task was approximately 5%. Hence, we confirmed that eight bipolar electrode pairs equally spaced around the forearm are dense enough to capture information regardless of sensor placement when acquiring sEMG signals provided that the same sensor placement is used during training and recognition.

3.3. Inter-day motion recognition

We obtained the inter-day recognition errors from leave-one-out cross-validation shown in Fig. 8 using both fixed and

Table 4

Average motion recognition errors (%) considering 28–30 days and five models using fixed and incremental (inc.) training data.

Subject	Training type	Sensor placement on first day		
		N	I	O
S01	Fixed	16.7 ± 5.2	17.2 ± 7.0	20.0 ± 9.7
	Inc.	12.3 ± 4.3	11.9 ± 4.9	11.5 ± 5.4
S02	Fixed	10.5 ± 2.7	12.1 ± 4.9	11.8 ± 5.2
	Inc.	7.9 ± 2.0	7.8 ± 2.9	7.4 ± 2.9
S03	Fixed	15.7 ± 3.7	16.7 ± 6.6	14.7 ± 6.0
	Inc.	11.7 ± 2.9	11.4 ± 3.9	10.9 ± 4.0
S04	Fixed	15.5 ± 3.6	15.7 ± 6.0	14.0 ± 5.8
	Inc.	11.4 ± 2.8	11.0 ± 3.7	10.4 ± 3.8
S05	Fixed	19.0 ± 6.6	17.5 ± 6.7	19.0 ± 9.3
	Inc.	14.2 ± 4.3	13.9 ± 5.2	13.3 ± 5.5

incremental training data. The solid lines and colored marks indicate the results obtained using incremental training data for five subjects. The average motion recognition errors across the 28–30 days and five models for each subject and training approach are listed in Table 4. Incremental training led to improved recognition performance for all subjects and suppressed recognition instability regardless of the sensor placement. A two-way repeated measures ANOVA for average recognition errors at the three sensor placements (neutral, inward rotation, and outward rotation) and two training approaches (fixed and incremental) showed statistically significant effects of the training approaches ($F(1, 24) = 158.3$, $p < 0.001$) but no significant effect of sensor placement ($F(2, 24) = 0.02$, $p = 0.980$). In addition, no significant interaction was observed between training approaches and sensor placements ($F(2, 24) = 0.129$, $p = 0.879$). Using incremental training data retrieved significantly superior results compared to using fixed training data (post-hoc analysis using Tukey test, $p < 0.001$).

The incremental training data retrieved more accurate recognition than the fixed training data. In particular, after day 15, the classifiers based on fixed training were increasingly inaccurate (dashed lines in Fig. 8), but those based on incremental training remained with stable performance. At a fixed sensor placement, a previous study [46] reported that LDA classifiers with optimal sEMG features obtained from clinical devices can offer robust motion recognition performances (6.6% error) during long-term (21 days) use for 11 upper limb motions: (i) wrist extension, (ii) wrist flexion, (iii) ulnar deviation, (iv) radial deviation, (v) forearm pronation, (vi) forearm supination, (vii) hand opening, (viii) hand closing, (ix) key gripping, (x) pincer gripping, and (xi) index finger pointing. However, the performances of both approaches may be drastically deteriorated over days when considering various sensor placements. We also confirmed that indiscriminate sensor reattachment catastrophically decreased recognition performance (see movie named `indiscriminate_reattachment`, whose link can be found

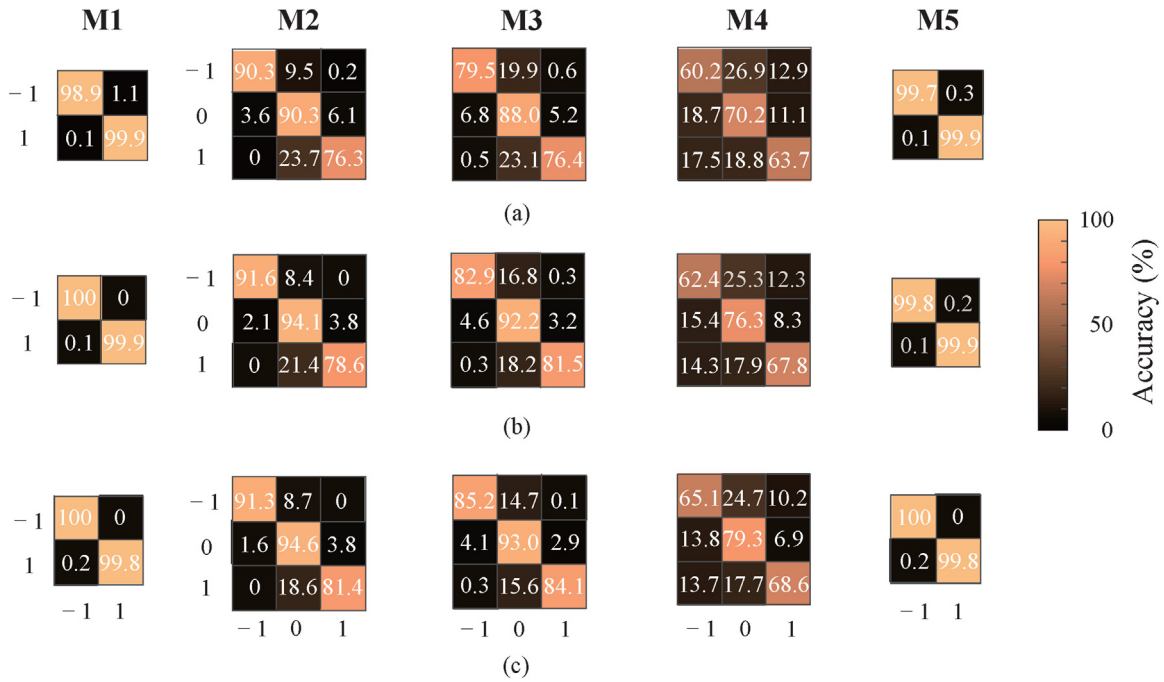


Fig. 7. Accumulated confusion matrices expressed in percentage for five recognition models over 30 days. Confusion matrices using (a) one, (b) two, and (c) three training trials for classification across subjects.

in file `Readme.md` from our GitHub repository). In these results, eight bipolar electrode pairs equally spaced around the forearm are not dense enough to capture information for suitable inter-day motion recognition regardless of whether the training approach was incremental or fixed.

To quantitatively assess the effect of changing sensor placement, we separated the average motion errors into nine patterns per training approach (Tables 5 and 6). A three-way repeated measures ANOVA for average recognition errors of the three sensor placements in the first-day data, three sensor placements for target-day data, and the two training approaches showed statistically significant effects of sensor placement on the target day ($F(2, 76) = 4.48$, $p = 0.0145$) and training approaches ($F(1, 76) = 47.32$, $p < 0.001$) but no significant effect of sensor placement on the first day ($F(2, 76) = 0.533$, $p = 0.589$). In addition, a significant interaction was observed between sensor placements on the first and target days ($F(4, 76) = 65.3$, $p < 0.001$), but no significant interaction between sensor placements was obtained on the first day and training approaches ($F(2, 76) = 5.07$, $p = 0.643$), and sensor placements on the target day and training approaches ($F(2, 76) = 0.605$, $p = 0.948$). When the sensor placement changed from I to O or vice versa, the performance under fixed training was significantly deteriorated (post-hoc analysis using Tukey test, $p < 0.05$).

Nevertheless, motion recognition errors always exceeded 5% over 30 days. In both training approaches, the results were affected by data depending on the sensor placement. Thus, we also verified the motion recognition errors using the information from each sensor placement (Fig. 9). The average motion recognition errors across 10 days and five models per subject and training approach are listed in Table 7, where the rows containing the fixed training results are the same as the columns containing the results from the three sensor placements ($N \rightarrow N$, $I \rightarrow I$, and $O \rightarrow O$) in Table 5, because their values only depend on the first-day data. As the average performances based on data of a sensor placement were mostly higher than those using all data of the three placements (see Tables 4 and 7), the relationship between training approaches and recognition performances has the same trend as that when using all data. A

Table 7

Average motion recognition errors (%) considering 10 days and five models using data of the same electrode and either fixed or incremental (inc.) training.

Subject	Training type	Sensor placement		
		N	I	O
S01	Fixed	14.9 ± 3.1	13.6 ± 3.3	15.6 ± 2.7
	Inc.	11.1 ± 1.8	10.6 ± 4.2	12.4 ± 2.1
S02	Fixed	9.3 ± 1.7	8.8 ± 1.0	10.5 ± 3.1
	Inc.	7.3 ± 1.3	6.6 ± 0.9	8.5 ± 3.9
S03	Fixed	13.8 ± 1.4	12.7 ± 1.8	13.7 ± 1.5
	Inc.	10.6 ± 1.0	10.0 ± 1.5	11.3 ± 1.0
S04	Fixed	14.7 ± 2.4	13.0 ± 1.0	11.6 ± 1.0
	Inc.	11.3 ± 1.6	10.5 ± 0.9	9.8 ± 0.9
S05	Fixed	17.4 ± 5.0	18.0 ± 4.8	15.0 ± 2.9
	Inc.	12.1 ± 2.2	11.5 ± 1.1	13.1 ± 2.3

three-way repeated measures ANOVA for the average recognition errors of the three sensor placements, two training approaches, and two data distributions (i.e., using one placement and the placement combination) showed statistically significant effects of the training approaches ($F(1, 50) = 219.8$, $p < 0.001$) and data distributions ($F(1, 50) = 5.14$, $p = 0.0277$) but no significant effect of sensor placements ($F(2, 50) = 0.526$, $p = 0.919$). In addition, no significant interactions were observed between training approaches and sensor placements ($F(2, 50) = 0.114$, $p = 0.982$), training approaches and data distributions ($F(1, 50) = 8.814$, $p = 0.237$), and sensor placements and data distributions ($F(2, 50) = 1.01$, $p = 0.850$). These results indicate that sensor placement did not significantly influence the error on the first day, but motion recognition performances can be improved if sEMG data are acquired from the same sensor placement every day. Furthermore, incremental training also improved the recognition performance over fixed training for any sensor placement (post-hoc analysis using Tukey test, $p < 0.03$). Overall, our sEMG dataset provides linear predictors with 5–20% errors over 30 days of data records.

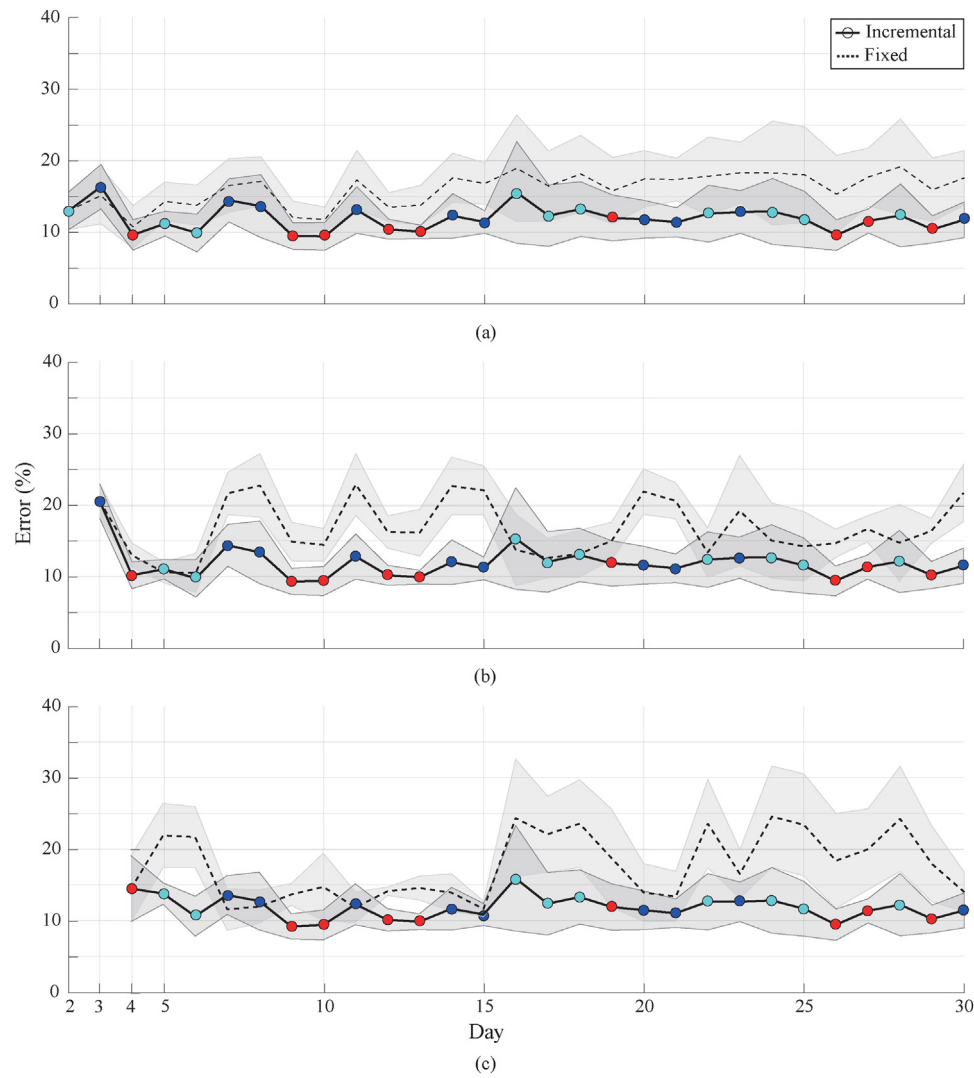


Fig. 8. Grand means and standard deviations of inter-day motion recognition errors over 28–30 days using fixed (dashed lines) and incremental (solid lines) training data. The first-day training data were acquired using (a) neutral, (b) inward rotation, and (c) outward rotation placements.

Table 5

Average motion recognition errors (%) under fixed training separated into nine patterns according to sensor placement (sub., subject).

Sub.	Sensor placement change (first day → target day)								
	$N \rightarrow N$	$N \rightarrow I$	$N \rightarrow O$	$I \rightarrow N$	$I \rightarrow I$	$I \rightarrow O$	$O \rightarrow N$	$O \rightarrow I$	$O \rightarrow O$
S01	14.9 ± 3.1	16.4 ± 4.9	20.1 ± 2.5	17.0 ± 3.0	13.6 ± 3.3	24.0 ± 3.0	20.5 ± 4.3	30.7 ± 3.2	15.6 ± 2.7
S02	9.3 ± 1.7	11.2 ± 1.7	11.9 ± 1.6	12.9 ± 1.4	8.8 ± 1.0	16.8 ± 2.7	11.6 ± 1.7	17.1 ± 1.2	10.5 ± 3.1
S03	13.8 ± 1.4	16.8 ± 1.4	17.9 ± 1.7	16.7 ± 1.8	12.7 ± 1.8	23.6 ± 1.6	14.2 ± 0.9	21.0 ± 1.4	13.7 ± 1.5
S04	14.7 ± 2.4	15.5 ± 1.3	17.8 ± 1.5	14.9 ± 1.8	13.0 ± 1.00	22.0 ± 1.8	15.5 ± 1.7	19.6 ± 1.6	11.6 ± 1.0
S05	17.4 ± 5.0	23.8 ± 6.3	17.6 ± 2.9	15.5 ± 1.3	18.0 ± 4.8	22.2 ± 5.0	19.9 ± 6.2	28.2 ± 3.9	15.0 ± 2.9

Table 6

Average motion recognition errors (%) under incremental training separated into nine patterns according to sensor placement (sub., subject).

Sub.	Sensor placement change (first day → target day)								
	$N \rightarrow N$	$N \rightarrow I$	$N \rightarrow O$	$I \rightarrow N$	$I \rightarrow I$	$I \rightarrow O$	$O \rightarrow N$	$O \rightarrow I$	$O \rightarrow O$
S01	10.4 ± 1.6	12.8 ± 4.8	14.6 ± 2.7	10.6 ± 1.6	12.7 ± 5.0	9.6 ± 2.5	11.7 ± 4.0	13.2 ± 5.2	13.5 ± 2.4
S02	7.3 ± 1.4	8.0 ± 1.0	9.1 ± 1.1	7.4 ± 1.3	8.0 ± 1.0	9.6 ± 2.5	7.5 ± 1.4	8.7 ± 1.7	8.6 ± 1.0
S03	10.6 ± 0.9	12.2 ± 1.6	13.3 ± 1.8	10.8 ± 0.9	11.9 ± 1.5	13.8 ± 3.1	11.1 ± 1.3	12.5 ± 2.0	12.7 ± 1.25
S04	10.7 ± 1.4	11.8 ± 1.4	12.6 ± 2.3	10.8 ± 1.3	11.5 ± 1.2	12.9 ± 3.2	11.1 ± 1.7	12.0 ± 1.3	11.7 ± 1.2
S05	12.0 ± 1.2	17.3 ± 3.3	14.6 ± 3.3	12.0 ± 1.2	17.5 ± 3.3	15.1 ± 4.0	12.4 ± 1.5	18.1 ± 2.6	13.8 ± 2.8

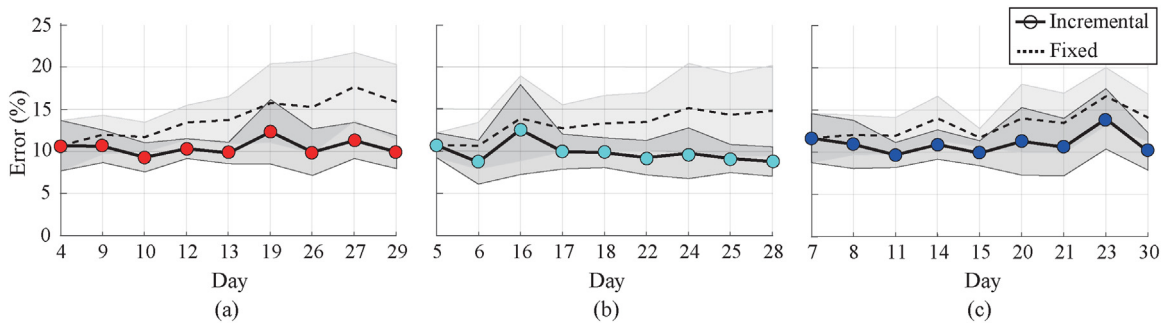


Fig. 9. Grand means and standard deviations of inter-day motion recognition errors over 10 days using fixed (dashed lines) and incremental (solid lines) training data at different placements: (a) neutral, (b) inward rotation, and (c) outward rotation.

4. Discussion

4.1. Data quality from Myo armband

The Myo armband sampling rate of 200 Hz has been considered very low [2,3,47], because MCSs usually require sEMG data sampled at over 1000 Hz during operation to capture enough frequency information. Li et al. [48] reported that a sampling rate below 400 Hz deteriorates the recognition performance. In addition, wearable devices capture noisier data than clinical devices because no paste or gel is used to reduce impedance during measurements. Thus, various problems remain to be solved regarding the use of armband sEMG devices, especially to improve reliability and minimize obtrusiveness [8,49–51].

The evaluation of our constructed dataset verified that armband sEMG devices can capture enough information within a day to suitably recognize combined forearm motions (Fig. 6). Poor data quality contributes to increasing error by noisy and undistinguishable features. However, recognition errors from every subject and day remained below 10% after accumulating training data, thus providing an acceptable classification accuracy for MCSs [41]. Therefore, the data quality of the employed armband sEMG device seems appropriate for accurate model construction when testing sEMG signals recorded at the same sensor placement as that used for training. Still, our dataset exhibits the common difficulties for distinguishing pronation and supination (Fig. 7), which remains challenging when using armband sEMG devices.

4.2. Sensor placement and sEMG data distribution

The histograms in Fig. 5 clearly show that the data distribution changes according to sensor placement. We recorded sEMG data at three different sensor placements with variations of 8 mm along the forearm, with positions I and O differing by 1.6 cm. This experimental design was based on previous studies [23,24]. The distributions for sEMG data obtained from each sensor placement moderately overlapped. However, changing the data distribution induced instability in motion recognition over multiple days (Fig. 8). In both fixed and incremental training, the errors ranged between 5 and 30%, despite using the same segmentation, feature extraction, and classification techniques. When recording sEMG data from the same sensor placement, motion recognition remained stable (Fig. 9 and Table 7). However, we cannot guarantee that the constructed models using sEMG data from the same sensor placement will provide completely stable performance over more extended time frames. Moreover, occasional model instability may be due to other factors changing sEMG data distribution, such as the physical and skin characteristics of subjects. Thus, armband device-based MCSs may not be stable for long-term use by employing conventional recognition algorithms. To overcome this limitation, an adaptation

approach is necessary for stable and safe MCS use during either short or long periods. As our dataset reflects such limitation and has information regarding various sensor placements, it may be effective to evaluate novel adaptation techniques and supervised learning techniques.

The recognition performance using incremental training (solid lines in Fig. 8) corroborated that this training approach is superior to fixed training. We assumed that combining data obtained from multiple placements deteriorated the recognition performance when using incremental training. However, this training approach was more effective than using fixed training in general. This result is counterintuitive to some extent, because the recognition models using training data from the neutral position should provide higher performance on all days than models using combined training data. Furthermore, equally spaced electrodes might easily deteriorate the overall information for recognition due to crosstalk [36]. Still, the armband sEMG device may provide stable recognition performance when the user perfectly repeats the electrode attachment to the same placement as that from the first day.

4.3. Signal processing module for motion recognition

Machine learning algorithms intend to provide accurate, fast, and stable motion recognition. The discriminability of our dataset was confirmed with a simple gold standard for sEMG processing using 11-dimensional features and linear models. Hence, some variations can be implemented for algorithm development and improvement.

In this paper, we used time-domain features combined with AR model coefficients with 250 ms analysis windows and 50 ms increments because the feature set and signal stream configuration are the gold standard and acceptable for real-time control of MCSs [34,35,52] although some studies use 200 ms analysis windows [12,53]. Phinyomark et al. [8] showed that time-domain features combined with AR model coefficients were most effective for recognizing sEMG signals by comparing eight sets of multiple features for myoelectric control based on four sEMG datasets containing finger, hand, wrist, and forearm motions of able-bodied subjects and amputees recorded with a wearable sEMG device of a sampling rate of 200 Hz. Furthermore, the loss of high-frequency components has a significant effect on nonlinear complexity features such as ZC and SSC. For accurately constructing classifiers using low sampling-rate data, which exhibit large intraclass variability and overlapping motion classes [52], extraction and selection techniques are essential to obtain high-quality, representative, and robust features. Smith and colleagues [54] discussed the optimal window length for sEMG-based motion recognition and the balance of the competing effects of recognition error and controller delay. They found the optimal window length that enables the best balanced perfor-

mance ranges from 150–250 ms. Thus, we used the feature and configuration for evaluation of sEMG-based motion classifiers in short/long-term use.

To construct a practical MCS in real environments, we designed five binary/ternary classifiers instead of a single multiclass classifier that recognizes multiple-DOF motions. Most of the sEMG-based motion recognition studies employ a single multiclass classifier [55,56]. Since the motions of the wrist and forearm are composed of simple DOF-wise actions defined by skeletal constraints, it is an effective option to prepare a classifier for each DOF. Young et al. [57] investigated a similar idea of preparing a classifier for each DOF for recognizing discrete and combined wrist and hand motions, though their three classification strategies include a separate classifier for detecting a specific discrete/combined motion without no motion class. The reason we used ternary, not binary, classifiers was to let the system be in an idle state, preventing jitter switching between opposite directions. Introducing an idle command has been tried in other biomedical interfaces like brain–computer interfaces [58]. Detecting rest from sEMG signals is easy, and we introduced a complete idling state in classifiers (Fig. 4, “Rest (–1)” for Model 1 and “Other (0)” for Models 2, 3, and 4).

Linear feature projection techniques, such as principal component analysis [59] and uncorrelated LDA [60], can be integrated into our recognition framework to enable fast sEMG processing. Improving the computation time is necessary because the sequential process in this study (i.e., input new data in windows of 250 ms, extract 88 dimensional features, and classify unknown data into five labels) implemented in C++, C#, and Unity took 263.2 ± 18.9 ms (five measurements), including the onset time of 250 ms for data preparation, on a PC running Microsoft Windows 10 with a 64-bit Intel® Core™ i7-7500K CPU at 2.70 GHz with 16.0 GB RAM.

4.4. Future work

To construct stable and robust classifiers, transfer learning has been used by including a calibration step before the reusing phase [21,24,25,61]. Such approach modifies the learned classifier adaptively to new data based on the calibration, as it assumes that calibration data are generated from the distribution of unseen data. The characteristics of our dataset would allow to investigate transfer learning regarding inter-placement and inter-subject transfer.

Moreover, the dataset may be valuable for testing new machine learning algorithms under realistic conditions of long-term use. However, it will be important to test any optimal algorithm developed with this dataset on various armband sEMG devices to validate their applicability.

4.5. Limitations

In this study, we only considered electrode shifts around the circumference of the forearm. It is also important to consider proximal and distal electrode shifts, which drastically affect the sEMG signals and consequently the MCS performance.

5. Conclusion

As armband sEMG devices have attracted considerable attention for developing MCSs in daily life environments, ensuring their reliable operation over time before deployment to the public is of paramount importance to ensure stability and safety during operation. This study confirmed the ability of recognizing intraday and inter-day motions with LDA classifiers by acquiring 22-class sEMG dataset of 30 recording sessions over 40–42 days with three different sensor placements. The results highlighted the robustness of intraday recognition and the difficulties regarding inter-day recognition. In addition, the results showed the effects of the various

sensor placements and training approaches as well as their interactions on the performances of motion recognition. This study and dataset represented a step forward to improve the evaluation of MCSs using sEMG signals of the forearm, as the diverse nature of the dataset allow us to quantitatively examine potential recognition algorithms for various applications.

Authors' contributions

Suguru Kanoga: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Atsunori Kanemura:** Conceptualization, Writing – original draft, Writing – review & editing. **Hideki Asoh:** Supervision, Funding acquisition.

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Conflicts of interest

The authors declare no conflicts of interest.

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